

COL761: DATA MINING · ASSIGNMENT 2 · QUESTION 1

k-Means Clustering in High Dimensions

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1 Elbow Detection: Method and Reasoning

For each $k \in \{1, \dots, 15\}$, k-means was run ten times with independent random initialisations and the best objective value

$$f(k) = \min_{\{C_j\}} \sum_{j=1}^k \sum_{x_i \in C_j} \|x_i - \mu_j\|^2$$

was retained. Selecting k from the resulting curve requires identifying the *elbow*: the point where adding another cluster yields diminishing returns. A naive second-difference rule $\Delta_k^2 = |f(k+1) - 2f(k) + f(k-1)|$ is sensitive to the absolute scale of the objective, which varies with n and d , making it unreliable across datasets.

Instead, the **normalised-distance (perpendicular) method** is used. Both axes are mapped to $[0, 1]$:

$$\tilde{k} = \frac{k - k_{\min}}{k_{\max} - k_{\min}}, \quad \tilde{f} = \frac{f(k) - f_{\min}}{f_{\max} - f_{\min}}.$$

Let $\hat{\ell}$ be the unit vector along the line connecting the first point $P_1 = (\tilde{k}_1, \tilde{f}_1)$ to the last $P_N = (\tilde{k}_N, \tilde{f}_N)$. For each k , define $\vec{v}(k) = (\tilde{k} - \tilde{k}_1, \tilde{f} - \tilde{f}_1)$. The perpendicular distance from the curve to ℓ is

$$d(k) = \|\vec{v}(k) - (\vec{v}(k) \cdot \hat{\ell}) \hat{\ell}\|,$$

and the optimal number of clusters is

$$k^* = \arg \max_k d(k).$$

Geometric Interpretation

k^* is the point on the normalised curve farthest from the diagonal—where the curve deviates most from a linear decrease. Because normalisation precedes the distance computation, the result is invariant to the magnitude of the objective, allowing consistent elbow identification across both datasets.

2 Objective Curves and Selected k

Figures 1 and 2 show the elbow curves for datasets 1 and 2 respectively.

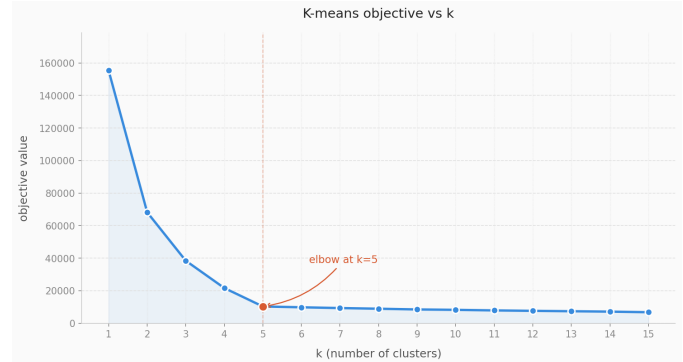


Figure 1. Dataset 1: objective vs. k , elbow at $k^* = 5$.

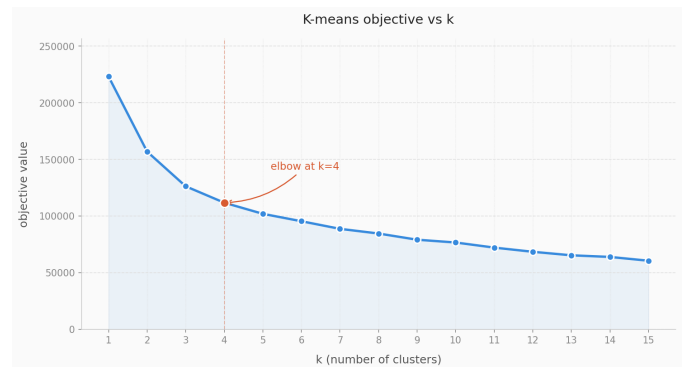


Figure 2. Dataset 2: objective vs. k , elbow at $k^* = 4$.

For dataset 1 the objective drops steeply through $k = 5$; beyond that, successive reductions are small, giving $k^* = 5$. For dataset 2 the sharpest bend occurs at $k = 4$, giving $k^* = 4$.

3 Assumptions Underlying k-Means

Structural Assumptions

- ▷ **Spherical and isotropic clusters.** The squared Euclidean objective implicitly assumes $\Sigma_j \propto I_d$ for every cluster j . Elongated or skewed clusters are poorly captured.
- ▷ **Comparable cluster sizes.** The centroid update averages all assigned points equally. Heavily imbalanced clusters cause large clusters to absorb nearby points that belong to small ones.
- ▷ **Linear separability.** Hard Voronoi assignment cannot recover non-convex or interleaved clusters regardless of k .
- ▷ **Homogeneous feature scales.** Because only ℓ_2 distance is used, high-variance features dominate; standardisation is required beforehand.

4 Limitations of Elbow-Based Model Selection

- ▷ **Subjective kink.** The curve often bends gradually; different automatic detectors can disagree by one or two positions.
- ▷ **No uncertainty estimate.** There is no significance test to distinguish a structural break from noise; different random seeds can shift the apparent elbow.
- ▷ **Local-optima contamination.** If k-means has not converged to the global minimum at some k^* , the inflated objective there shifts the elbow erroneously, even with multiple restarts.
- ▷ **Aggregation bias.** The scalar objective masks per-cluster behaviour; if clusters have very different densities the elbow reflects an average effect rather than a true structural change.

5 Improvements and Alternatives

- ▷ **Gap statistic [1].** Compares observed scatter to a uniform-null reference, providing a statistically grounded stopping rule that correctly handles the $k = 1$ case.
- ▷ **Silhouette coefficient [2].** Measures how well each point fits its own cluster versus the nearest other; the mean over all points peaks at the best k and lies in $[-1, 1]$, making it directly interpretable.
- ▷ **k-means++ [3].** Initialises centroids with probability proportional to squared distance from already-chosen ones, reducing run-to-run variance and the local-optima problem.
- ▷ **Stability selection.** Clustering multiple subsamples and measuring agreement via the adjusted Rand index rewards k values that produce consistent partitions, independently of objective scale.

6 When k-Means Fails and How to Detect It

Failure Modes

- ▷ **Non-convex geometry.** Two concentric rings in $\mathbb{R}^2$ cannot be separated by any Voronoi partition; k-means cuts through each ring rather than isolating them.
- ▷ **Anisotropic clusters.** Elongated Gaussians aligned along the same axis are often split internally rather than separated, because the objective penalises elongation.
- ▷ **Density imbalance.** A dense cluster of 1000 points overwhelms a sparse cluster of 10, absorbing its members into the dominant group.
- ▷ **High-dimensional noise.** In $d \gg 3$ with many irrelevant features, concentration of measure makes all pairwise distances nearly equal [4], rendering nearest-centroid assignment essentially random.

Detection procedures.

- ▷ **Hopkins statistic.** Values near 0.5 indicate no clustering tendency; be cautious if $H < 0.75$.
- ▷ **UMAP / t-SNE projection.** A 2-D scatter reveals non-convex structure or density imbalance before clustering is run.
- ▷ **Low silhouette across all k .** Mean silhouette below 0.2 suggests the spherical-cluster assumption is violated; a full-covariance GMM selected by BIC should be preferred.

References

- [1] R. Tibshirani, G. Walther, T. Hastie, "Estimating the number of clusters via the gap statistic," *JRSS-B*, 63(2):411–423, 2001.
- [2] P. J. Rousseeuw, "Silhouettes: a graphical aid to cluster validation," *J. Comput. Appl. Math.*, 20:53–65, 1987.
- [3] D. Arthur, S. Vassilvitskii, "k-means++: The advantages of careful seeding," *SODA*, pp. 1027–1035, 2007.
- [4] K. Beyer et al., "When is 'nearest neighbor' meaningful?," *ICDT*, LNCS 1540, pp. 217–235, 1999.